Organization View.

The making of a world-leading 'AI' service

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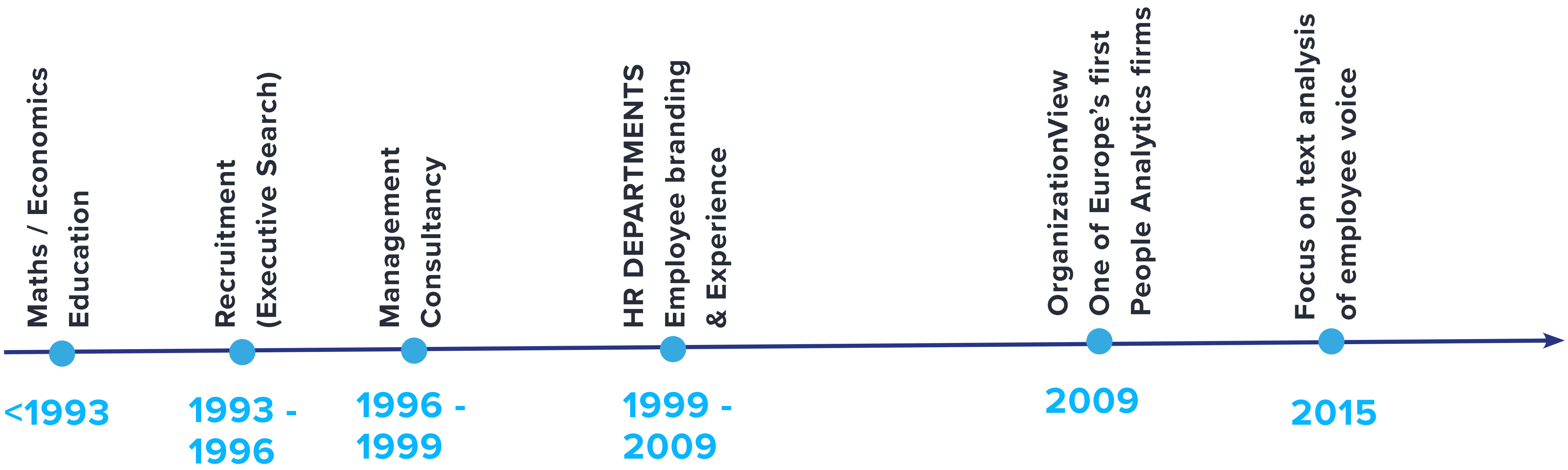
Helping employees be heard and leadership hear what is really happening

Background

From Seventies programming nerd to expert on employee text



Gartner Cool Vendor 2018 ™

WINNER PEOPLE ANALYTICS AWARDS 2016 EMERGING GAME-CHANGER

Problem we're solving

Multi-language analysis of large volumes of employee ideas.

- Understanding employees needs and perceptions is essential for improving employee experience
- Employees can be a key source of ideas, innovation and business improvement
- Rich text data is the best way of capturing these thoughts
- ...But has historically been hard to analyse at scale
- **"Would it be possible to make analysis of text data as easy as quantitive data?"**

Text analysis as quick, often exploratory analysis

Quantitative Analysis

Explains what happens / could happen (when, to what etc.)

Time consuming

Do you have the right and enough data?

Qualitative Analysis

Explains why something is happening

Time consuming, but don't need as many subjects

Effective at driving action

Does it scale?

Text Analysis

Can indicate both what & why

Quick & 'cheap'

Doesn't need a pre-built 'hypothesis'

Can usually find existing data sources

Good at showing linkages

Short distance to detail

Accuracy and performance

We aim to have the highest performing text analysis on the market

250 common themes found in employee comments identified in the base model.

Less than 1% Misclassification rate

Abbreviations and acronyms identified, linked to a dictionary of 2,000 definitions

Each theme clustered to identify different ways that employees describe the topic

‘Quotable’ examples identified and highlighted

Proper nouns (names, places, competitors, software products) identified & available for reporting

"AI" IS AUTOMATION ON STEROIDS

(FIRST IDENTIFY WHAT YOU WANT TO AUTOMATE)

APPROACH 1: Grounded Theory Research

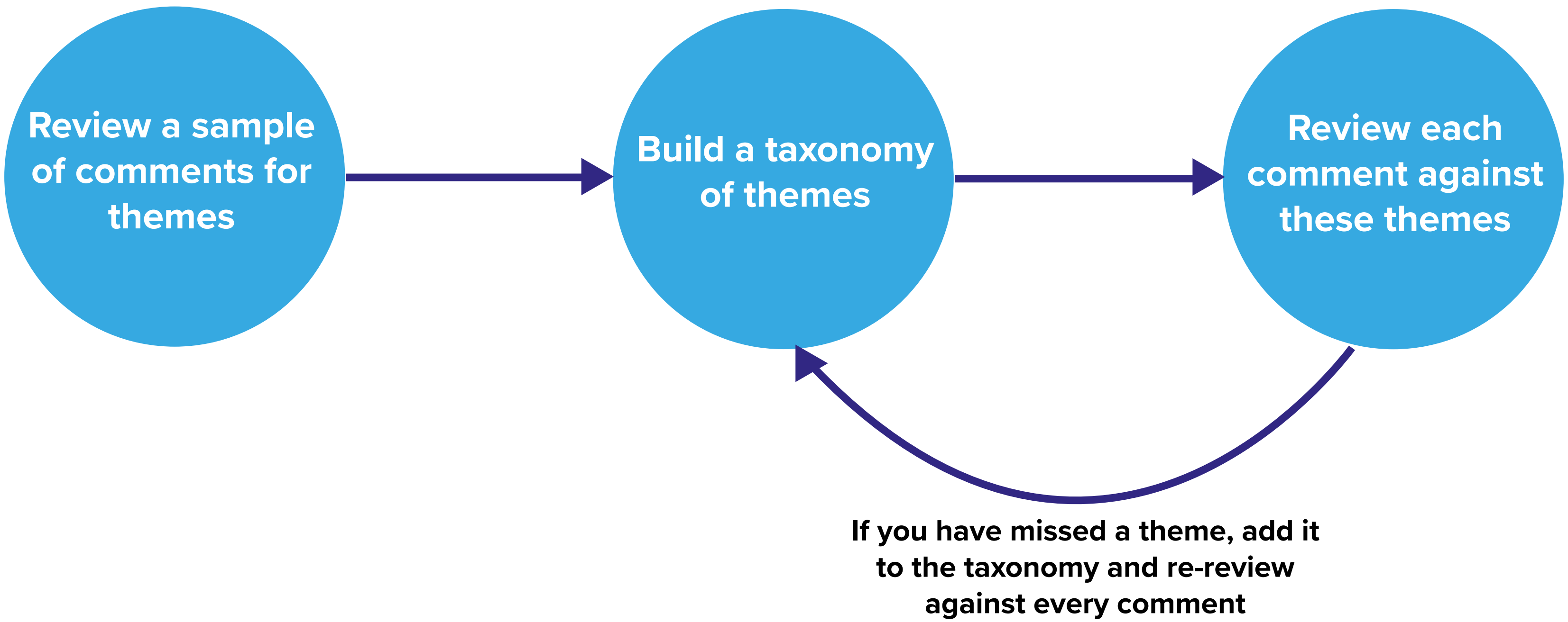
In Grounded theory we don't want to start with a framework, which might create bias in our interpretation. So we do a detailed 'line-by-line' study. We might do this before thinking about the question to be answered.

We view this as the 'gold standard'

Topics should be understandable in their own right. So not "Career Development" but instead "Lack of available opportunities"

APPROACH 2: Sample, identify, review

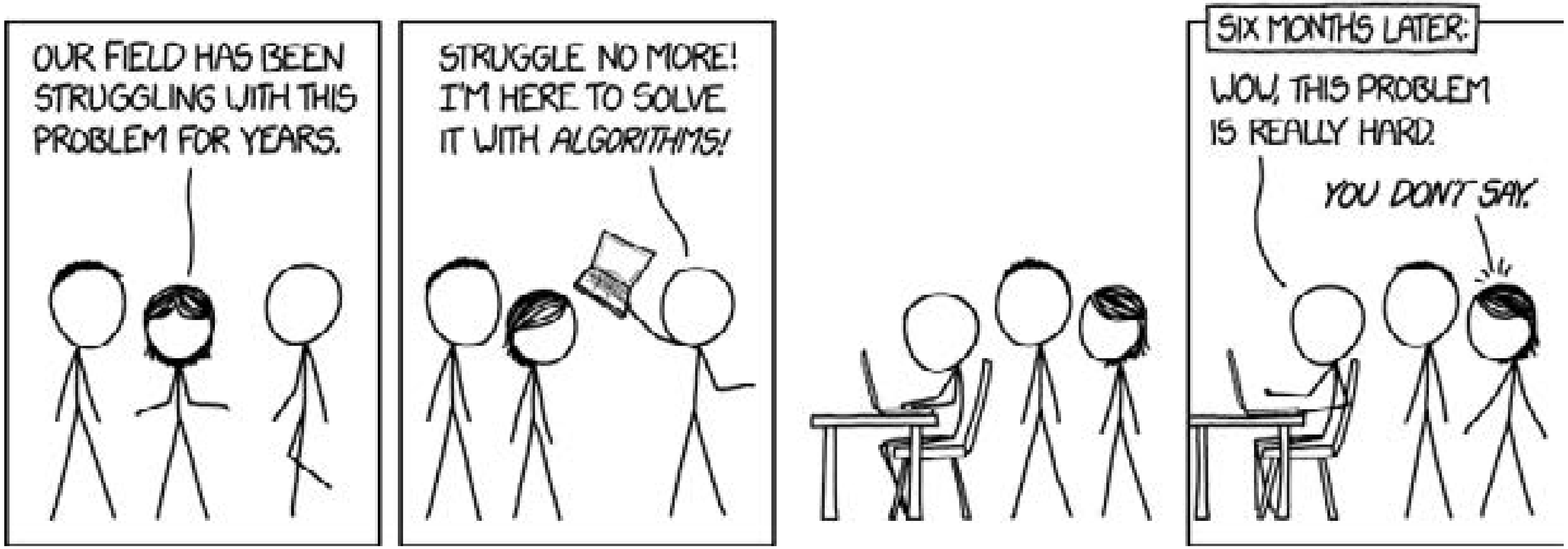
This approach has typically been used for large volumes of comments. The reduction in quality is balanced by increased speed & potentially lower skill requirements



KEY TECHNICAL CHALLENGES
a.k.a
“HELLO THEORY, MEET REALITY”

There are no easy solutions - everything is in the long tail

But we're getting good at solving specific problems



Need to understand the semantic meaning

"If you could do one thing to improve working for COMPANY what would it be?"

"more money, less hours!"

"Not being salary capped after 30 + years of loyal service"

"Competitive salary increases for internal movement, low pay rises encourage outward moves rather than talent retention"

"Better pay"

"Better wages"

"if I could I would give a pay rise to the living wage amount and not just the minimum wage, we do work hard for our company."

Need to understand the semantic meaning

“If you could do one thing to improve working for COMPANY what would it be?”

“more money, less hours!”

“Not being salary capped after 30 + years of loyal service”

“Competitive salary increases for internal movement, low pay rises encourage outward moves rather than talent retention”

“Better pay”

“Ha Ha. A raise”

“Better wages”

“if I could I would give a pay rise to the living wage amount and not just the minimum wage, we do work hard for our company.”

Context is vital for understanding text information

Generic text algorithms fail to provide contextual meaning to comments. We look at text in 3 ways:

The context of the Organization

The context of the Question

The context of the sentence structure

Context of the organization is highly beneficial

Employees will often use names rather than descriptions. They assume you know these.

Bangalore	**The offshoring centre**
Zürich	**The headquarters**
Jane	**The CEO**
‘Ask Anita’	**The internal chatbot**
Eagle	**The finance transformation programme**

Bias needs to be monitored & dealt with

Large language models can ‘impose’ their own interpretations of the data.

“A really French working style”

Theme: “Bureaucracy”

Text analysis historically expensive

HR has been trying to avoid capturing and analysing text.

POSITIVES	NEGATIVES
Contains richness & Granularity	Messy - typos, multiple-wordings
Enables people to add what they need / want to say	Introduction of subjectivity - potential bias
Easier to design a good UI	Risk of litigation
	Difficult to report

TECHNICAL CHALLENGES & APPROACH

Text classification requires manually created training data

There is no getting away from needing good training data

Most predictive models in HR:

There is access to a dataset which shows what you want to predict that the model can learn from.

Label	*Gender*	*Age*	*Location*
1	M	32	DE
0	F	25	UK
0	M	58	ES

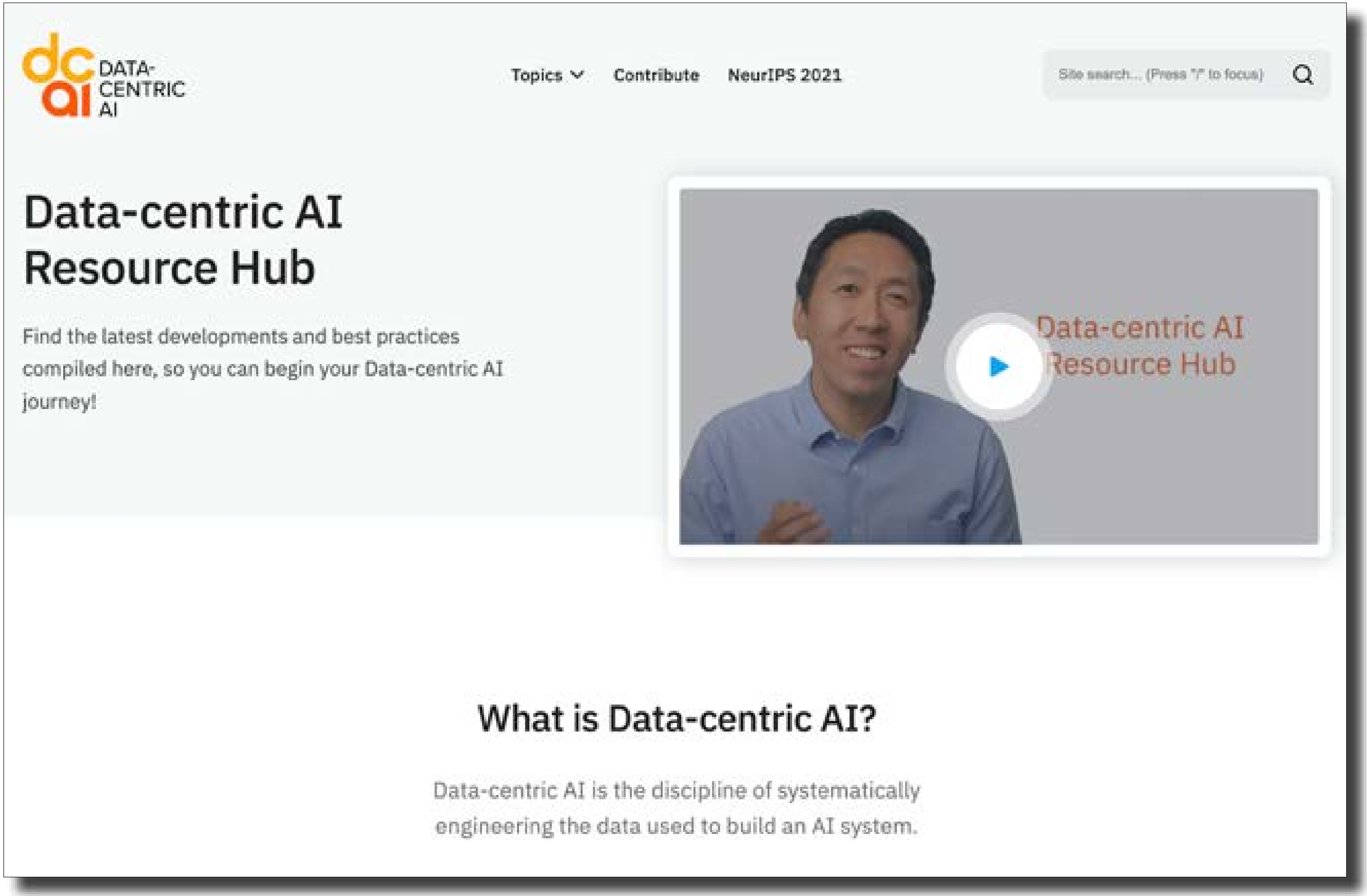
Text classification

No model exists - you have to create one!

Theme	*Sentence*
Pay	Can you increase my wages
Training	After the grad scheme there is no training available
Manager	My manager never provides training
Training	My manager never provides training

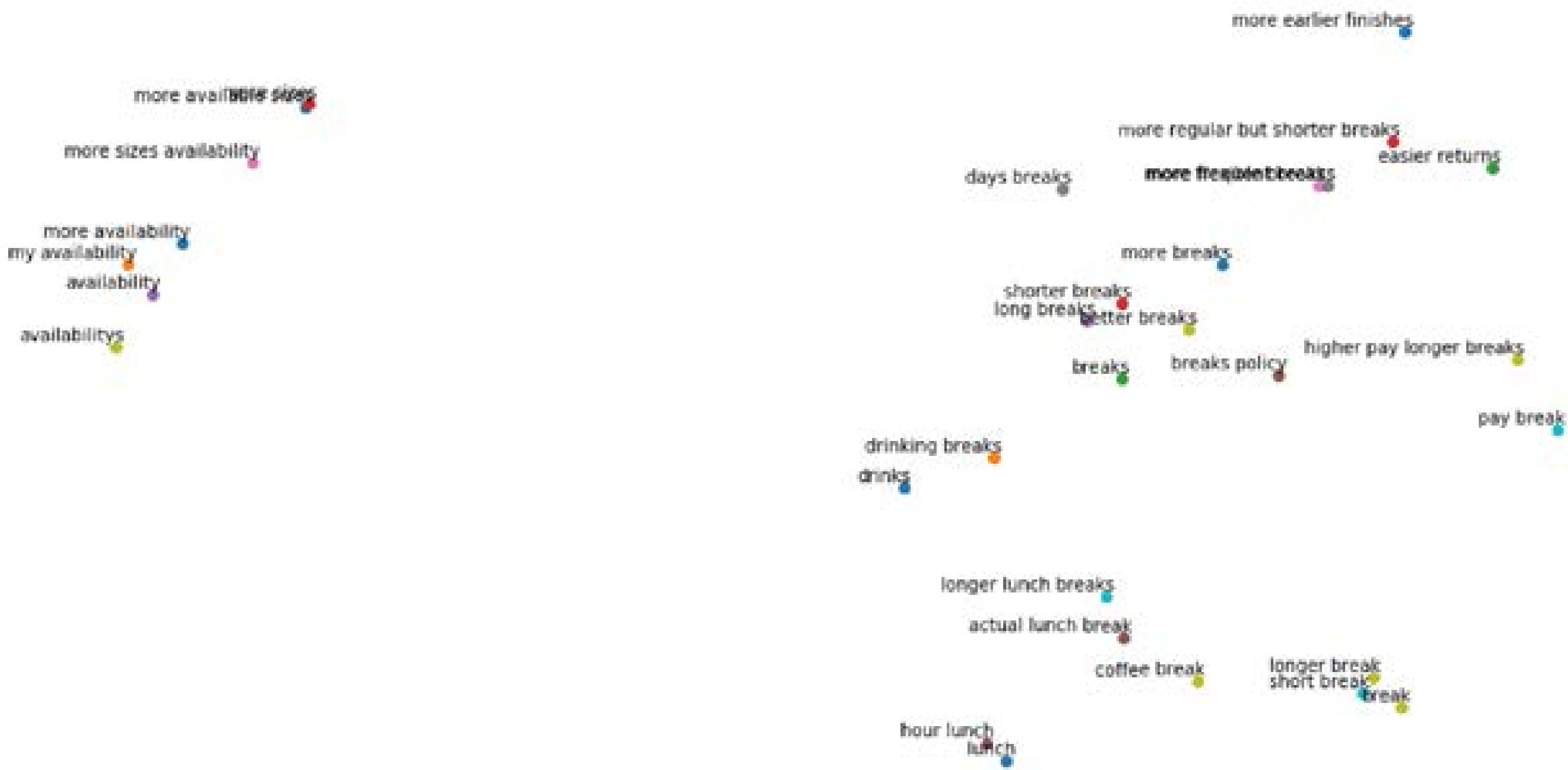
Data Centric AI - focusing on the training data not the model

Firms like Tesla often have 50x more people working on creating training data than building AI models.



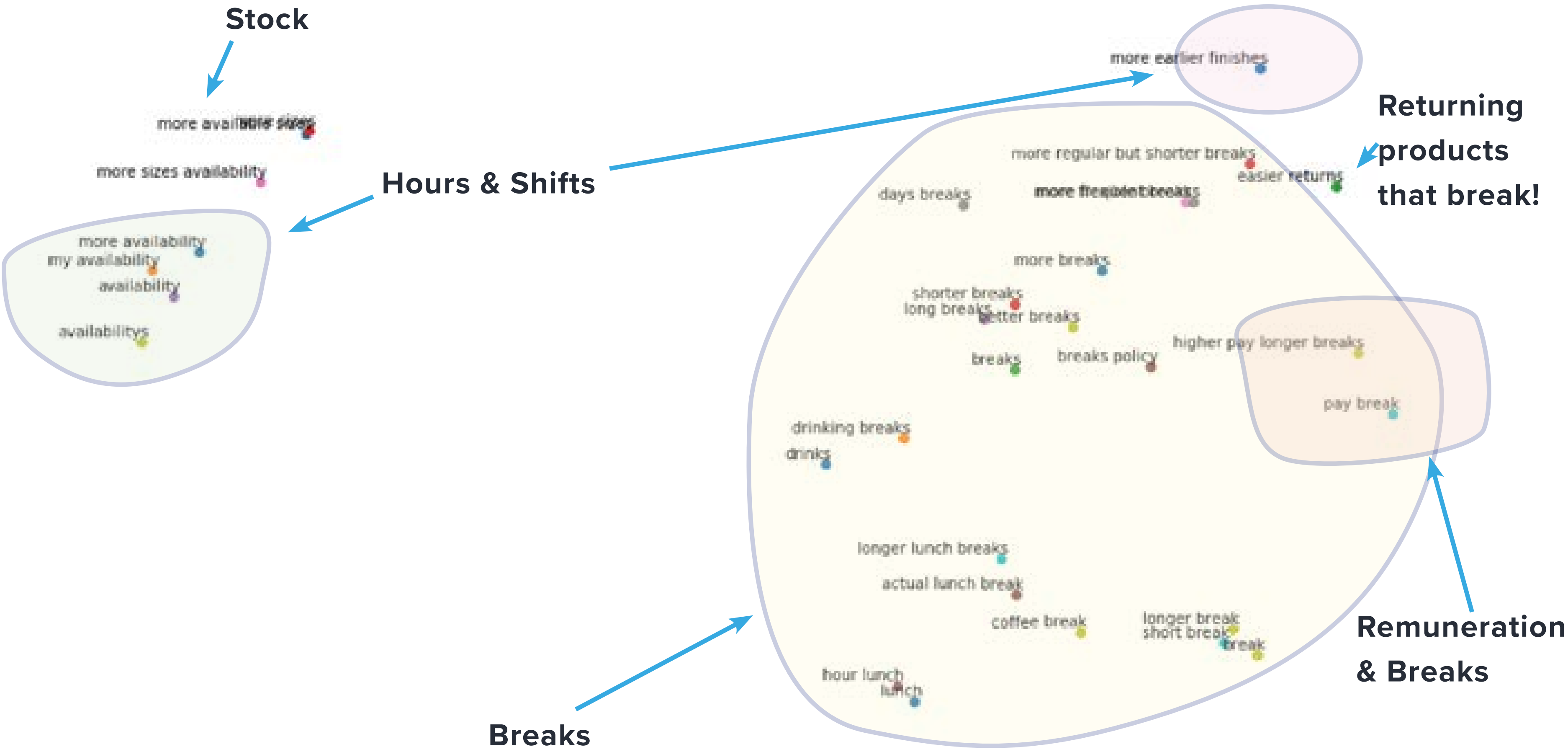
Finding semantically similar short phrases

Dimensionality reduction can give you an idea of what is happening



Finding semantically similar short phrases

Setting boundaries around themes to categorise ideas



Finding the edge-cases

Understanding your algorithm and where it's struggling is vital to getting really high performance.

How do we separate....	from this
my **workplace** (the place I work)	**Workplace**, the software from Facebook
an employee's **benefits** package	how an employee **benefits** from....
more **pay**	**pay** more attention
better **health**	better **health** insurance
Moral (most common misspelling of Morale)	**Moral** (ethics)

Labelling consistency / Inter-reviewer agreement

It's vitally important that labellers have a clear set of written guidelines for reviewers

Themes should be mutually exclusive

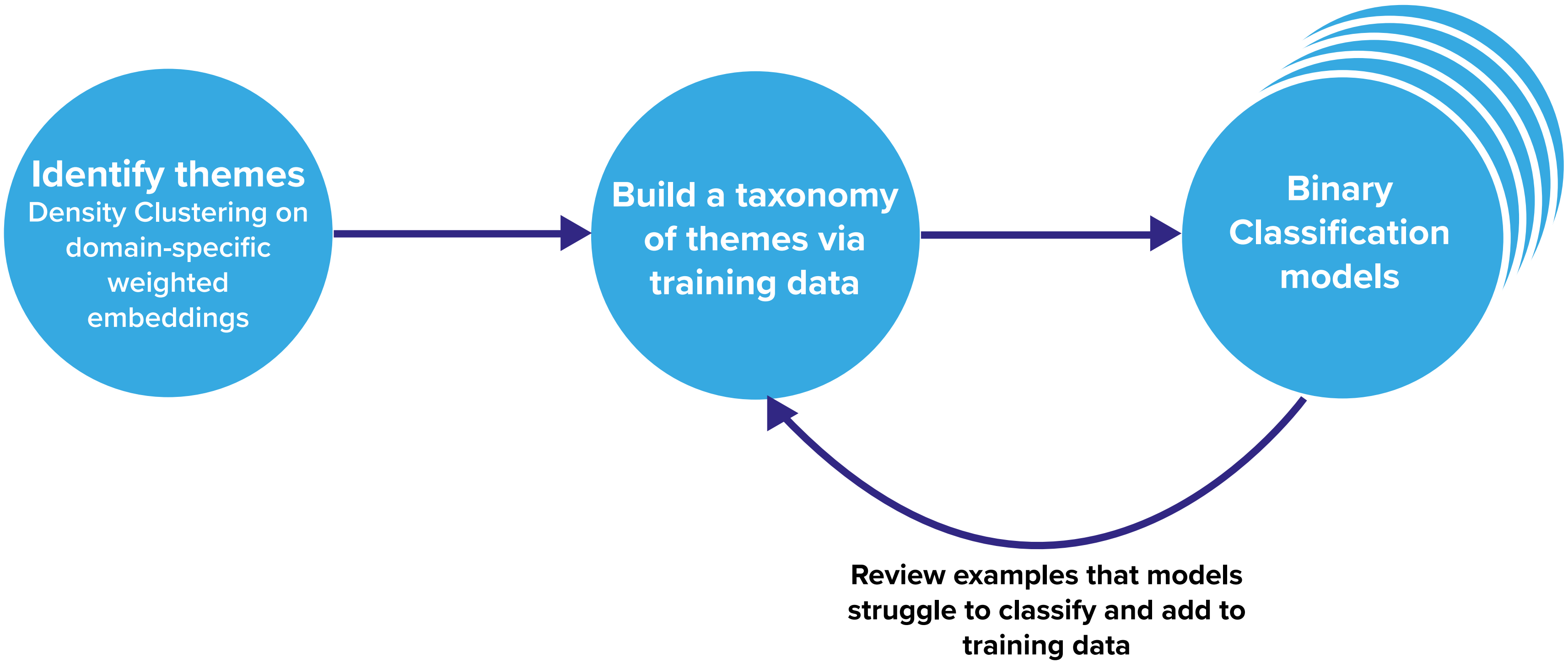
Themes should be objective

You need a process for managing theme evolution

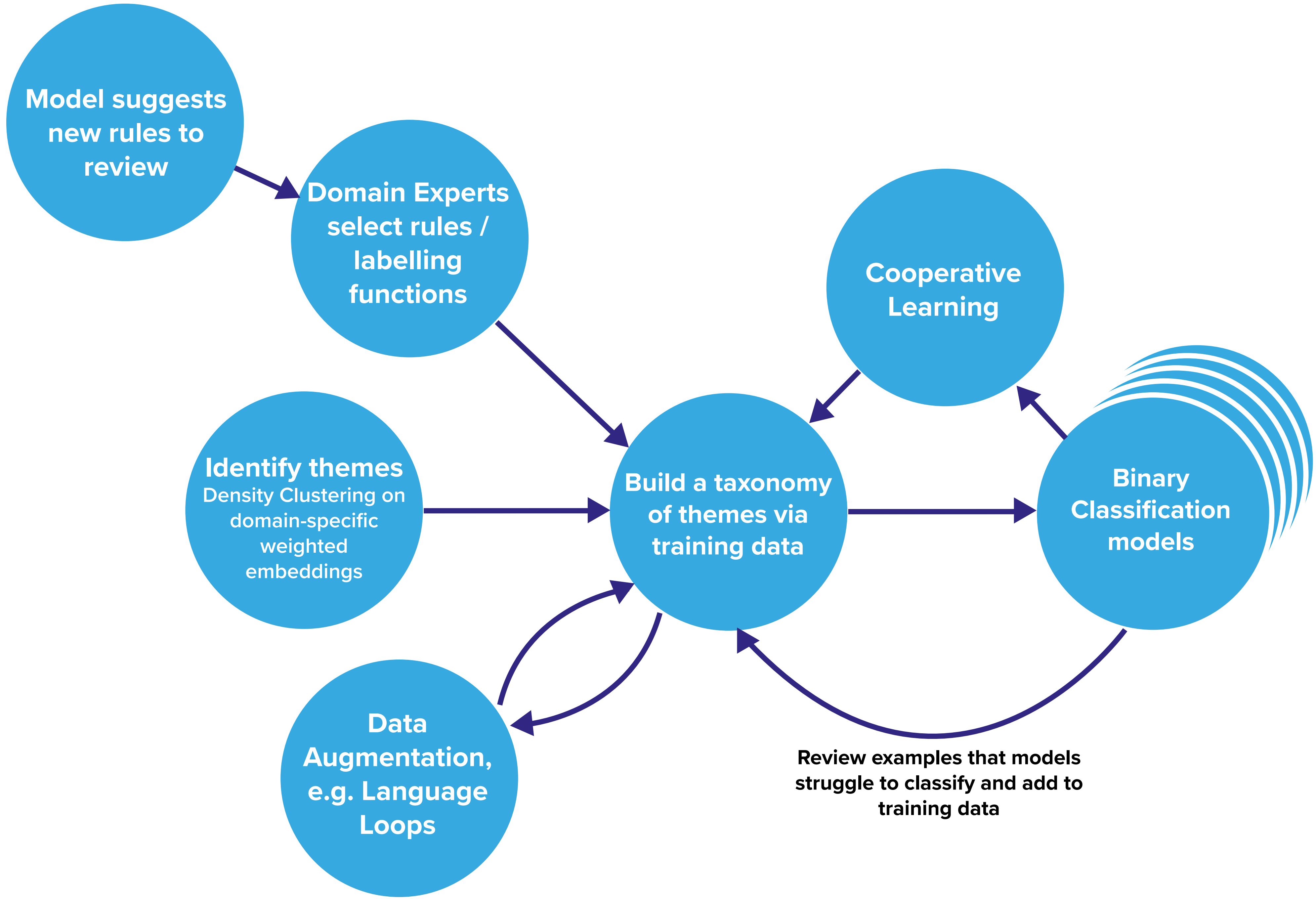
ML ASSISTED ANALYSIS

Building an inductive model with ML assistance

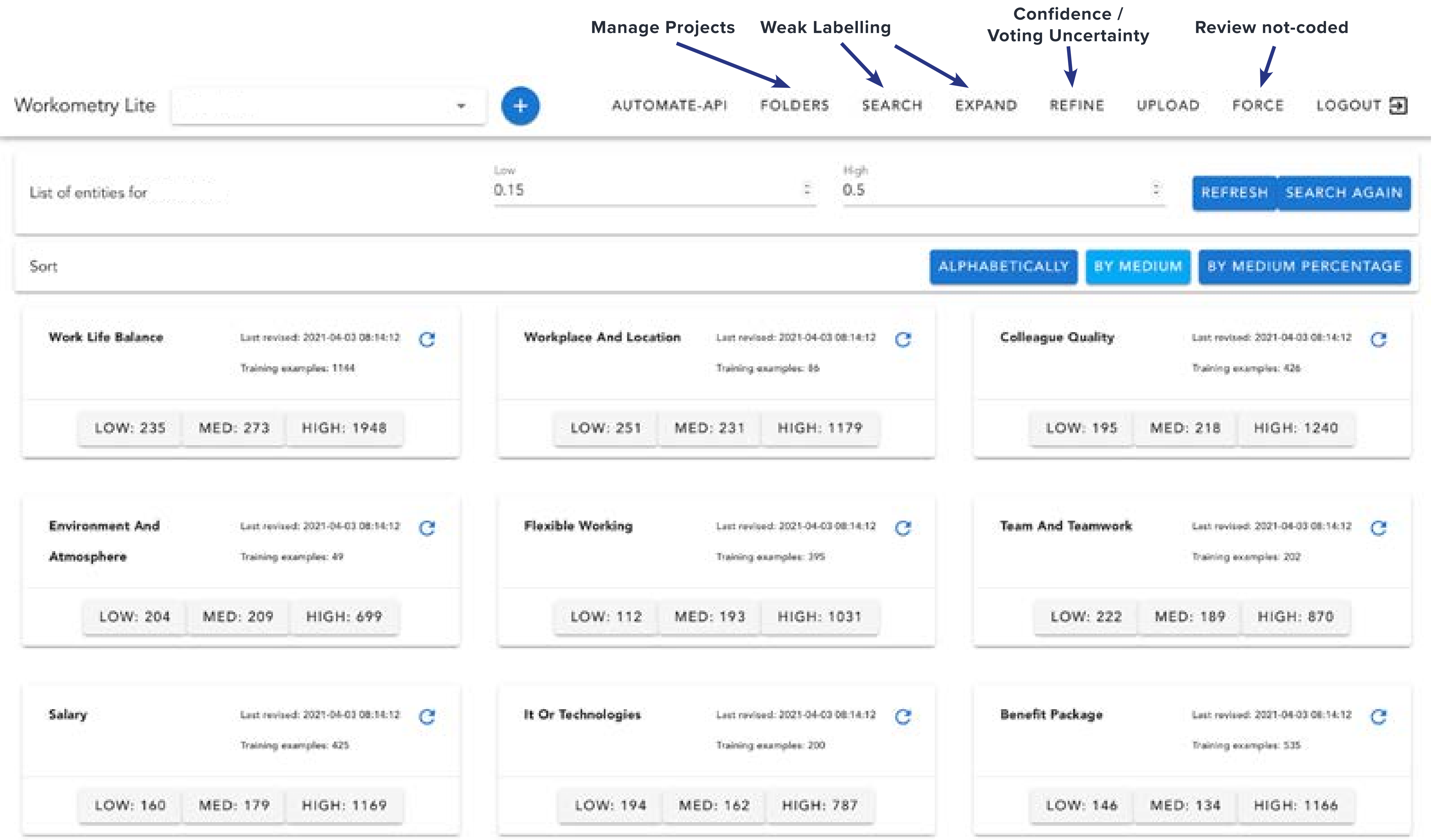
Our initial process was very similar



Of course the end solution is a lot more sophisticated

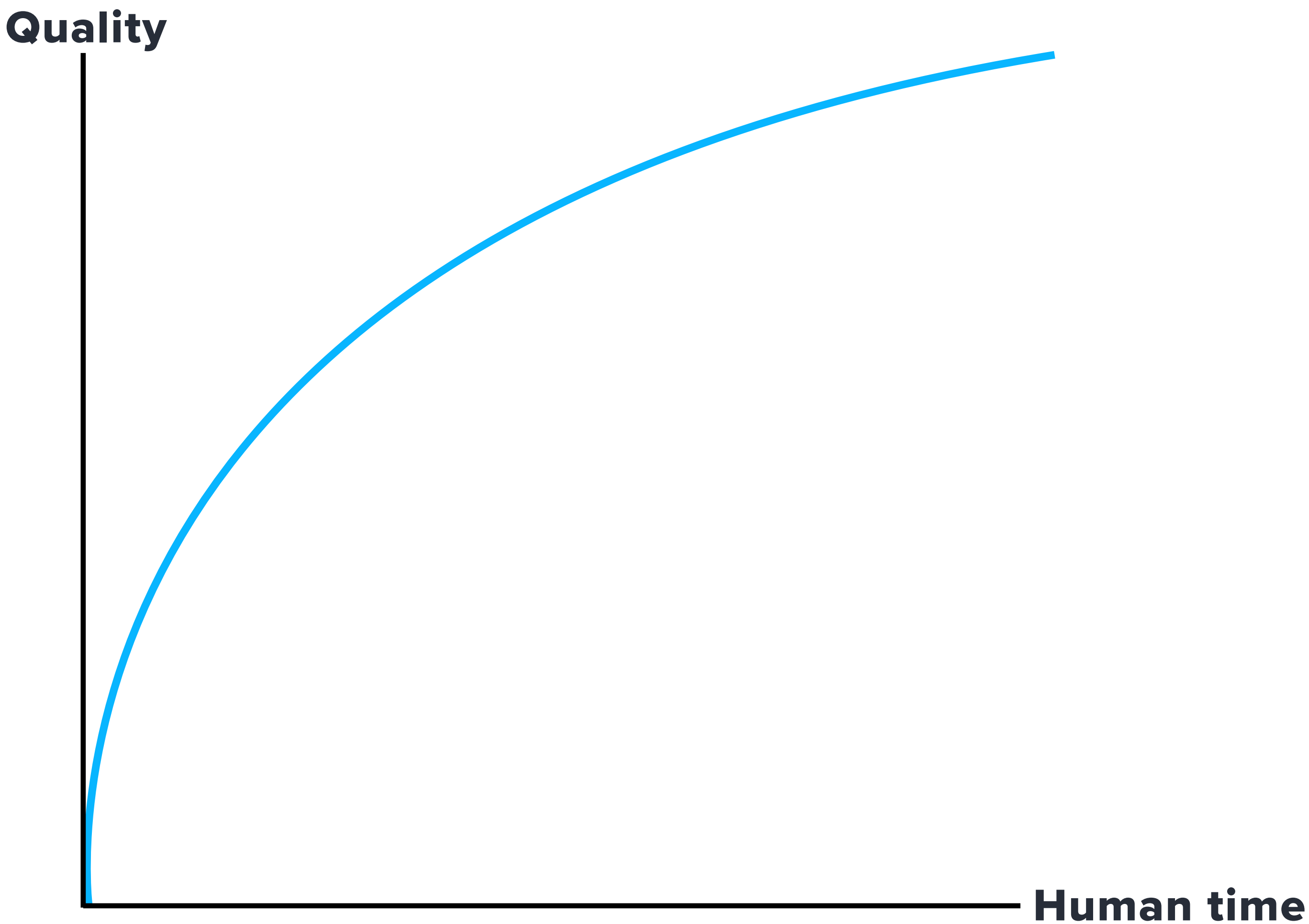


Internal tooling to enable Domain Experts to build & curate models



Optimising on two variables

Given our constraints, there are two things that define our success



How we structure our team

Our key focus is on industrialising this process to be quicker, with lower errors and cheaper

Coding and Curation team

Builds the models by editing training data

Infrastructure team

Develops WorkometryCI, our internal coding platform

Data Science team

Develops algorithms, tests ideas, monitors performance

CLASSIFYING TEXT ISN’T THE END GOAL

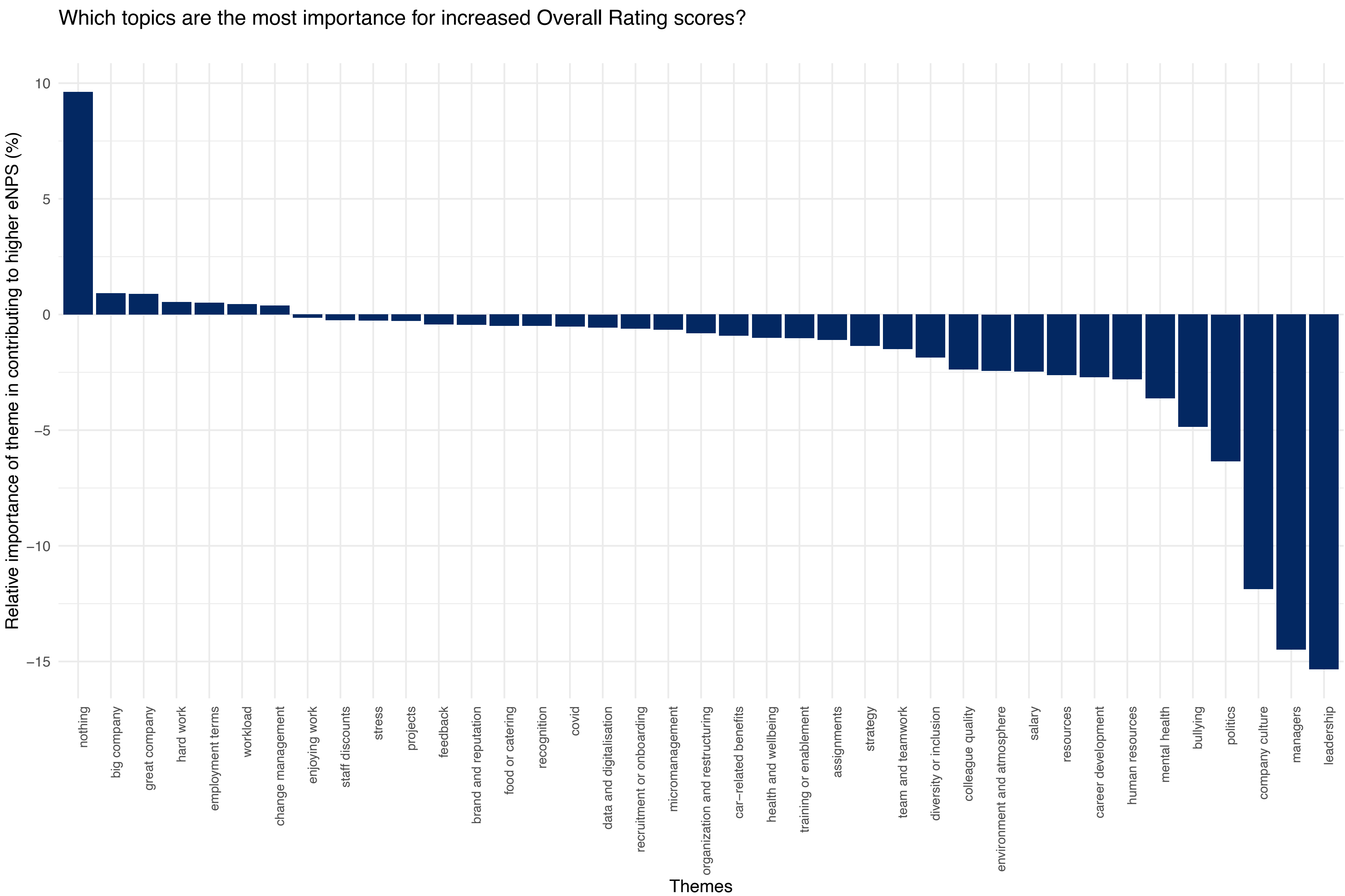
Helping answer executives key questions

After coding the text the real work starts. Post coding we run numerous machine learning models to spot patterns, identify supplementary information and provide context to the survey comments.

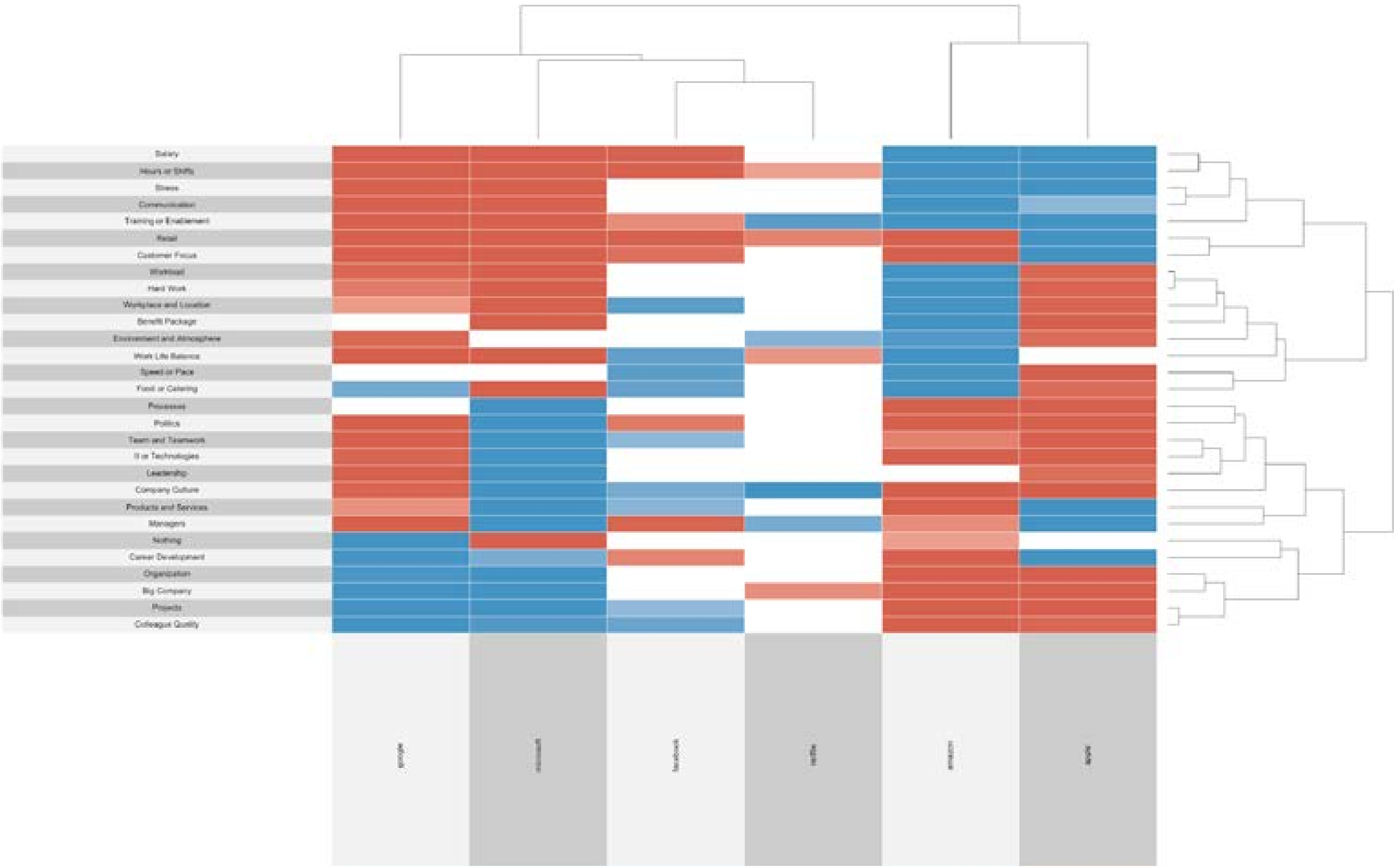
What are people saying?	Which groups are mentioning key themes?	Which themes are driving key metrics?
What do people mean by "X"?	What's a good example / quote?	What is the relationship between themes?

What is driving the thing you care about?

We can identify which themes are most important for 'driving' scale questions.



Pattern-spotting: Which groups talk about which topics?

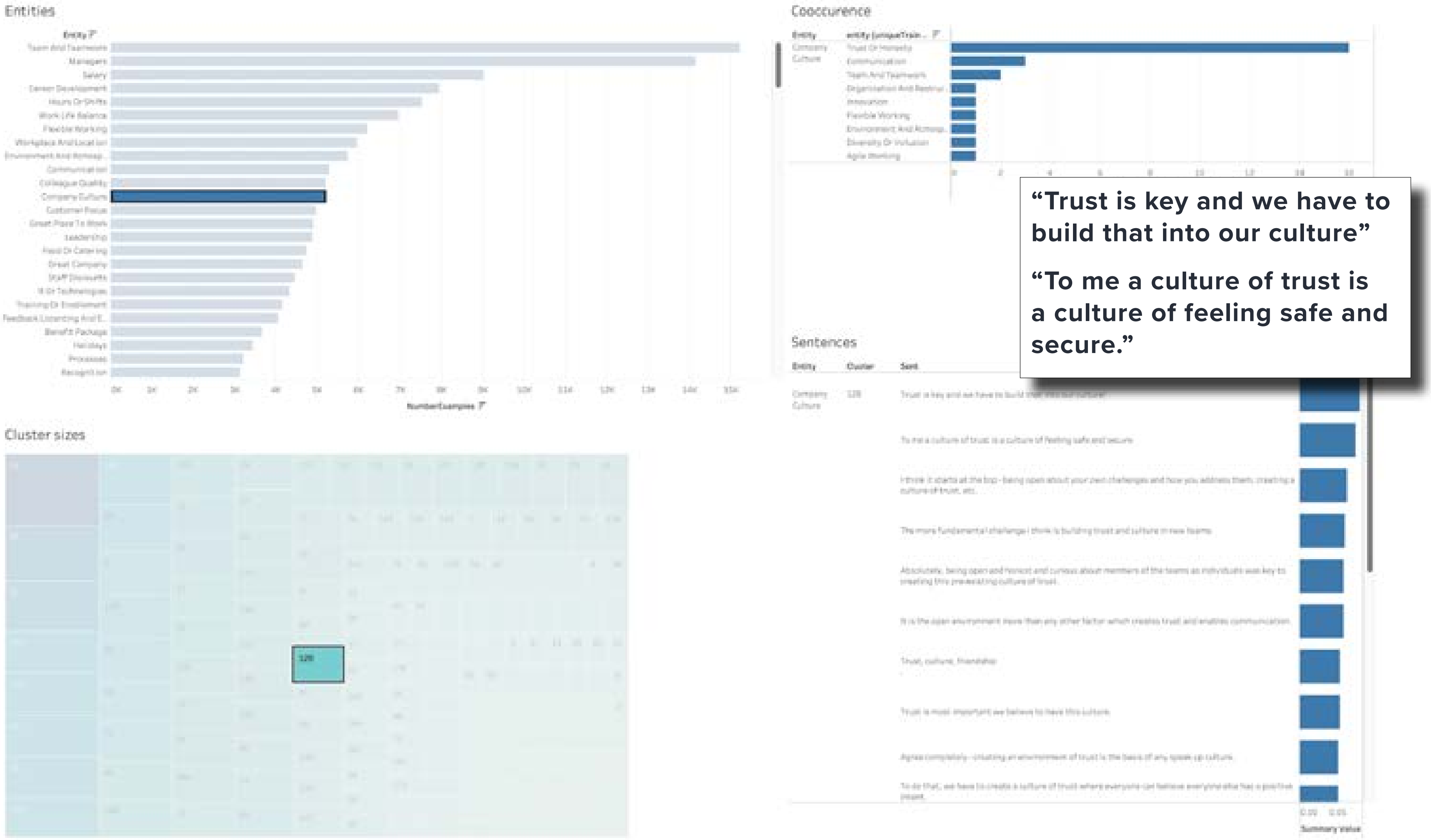


Identifying the different ways that people describe each theme

Biggest category is a good and flexible working culture

Grouping sentences together enables rapid understanding of messages

Whereas this one talks about how trust is important to build the culture



Summarisation

Text can be summarised, either by identifying key sentences (extractive) or re-writing the text (abstractive)

"Even when reported to management, the care factor seems to be non existent"

"There is no coherent discourse on the part of senior management regarding trust with subordinates."

"Thinking about the training and preparation of the workers is of no use, only that the management system is completed."

"In middle management, the health of the employee is not important"

"No communication from senior management on any issues."

"No text governing senior management exists within this division."

"Employees time is considered to have no value to management"

"There has been no accountability for senior management on production performance."

"There is no accountability at the management level and this filters down to the workers."

"As a rule, employee management is not part of the training in a university degree, but it is in lower management (master for several hundred hours)."

"Management are detached from employees."

"No respect from top level management None feels like upper management care about employees."

There is no coherent discourse on the part of senior management regarding trust with subordinates. There is no accountability at the management level and this filters down to employees. Management are detached from employees.

Using the verbatims as a knowledge base to answer new questions

We use Natural Language Understanding to help answer other important questions. Here the employees were asked 'what could improve working for [company]?'. Within those answers is valuable information to a variety of other questions.

QUESTION: Why do employees leave?

Reason	Answer provided
management decisions	Many of the employees who got rebadged to [Division] left the company in less than a year because of MANAGEMENT DECISIONS and none of them got a retained
lack of promotions and growth	There is a large populations of people leaving and LACK OF PROMOTIONS AND GROWTH are the reason it,
she did not get the support she needed	My coworker ended up leaving [company] because SHE DID NOT GET THE SUPPORT SHE NEEDED
to get a promotion	A lot of employees leave the company after a few years in order TO GET A PROMOTION and then return to [company].
they are not recognized	Even qualified older employees are left out and many left the company due to the reason THEY ARE NOT RECOGNIZED, on the contrary, they only know that they are charged day by day, with a huge workload.
workload is very heavy	High employee turnover, probably because WORKLOAD IS VERY HEAVY.
convoluted internal processes and excessive red tape	High employee turnover as new hires get demotivated very quickly by CONVOLUTED INTERNAL PROCESSES AND EXCESSIVE RED TAPE

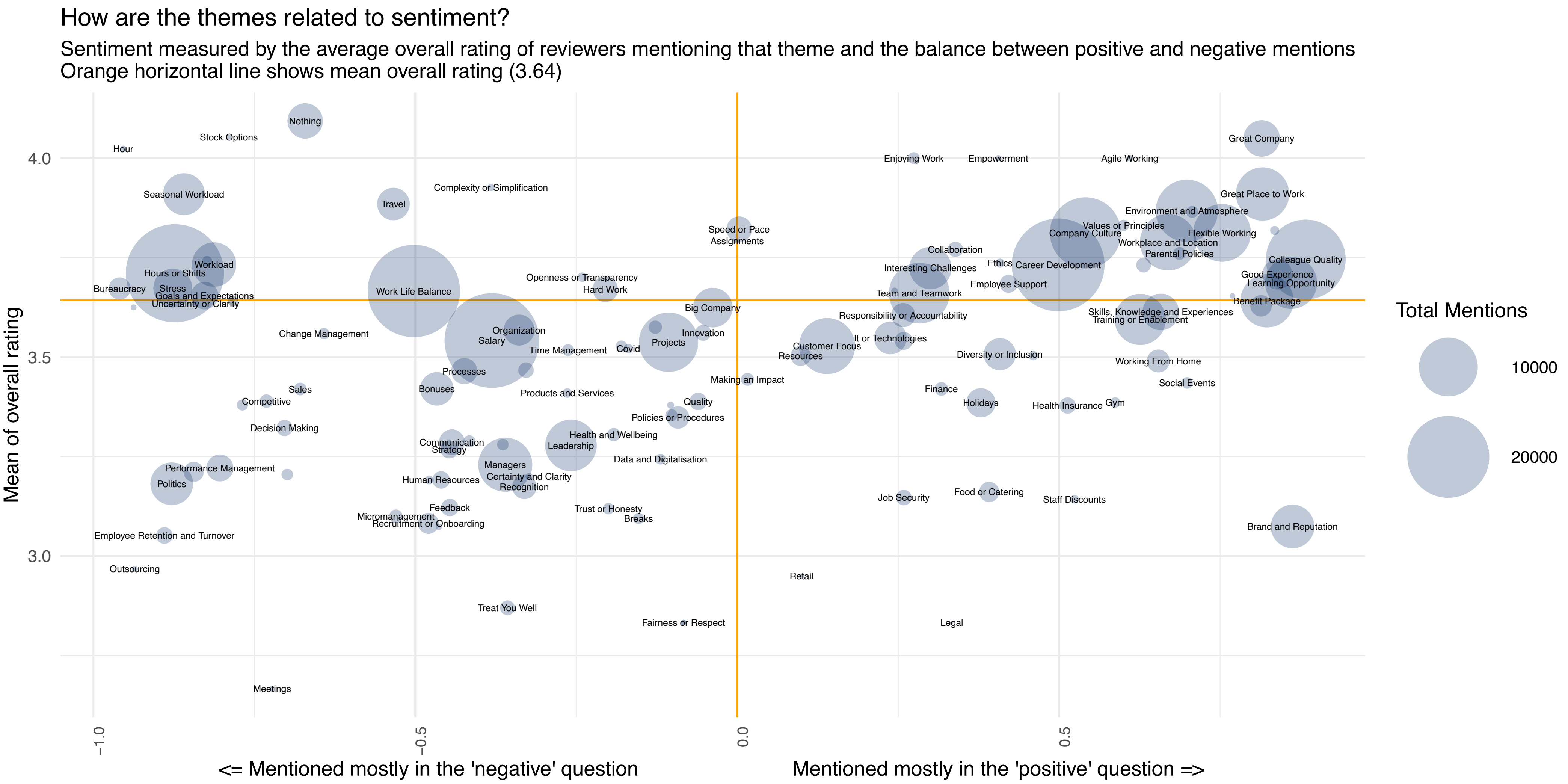
What is the employee trying to convey?

Understanding the answer type makes a big difference how you action / review it

EXPERIENCES	SUGGESTIONS	LINKED STATEMENT
"Last summer there was a huge vocal presence from senior leadership about diversity and representation."	"Make a commitment to neutralize the gender wage gap."	"If you really value diversity, put your money where your mouth is."

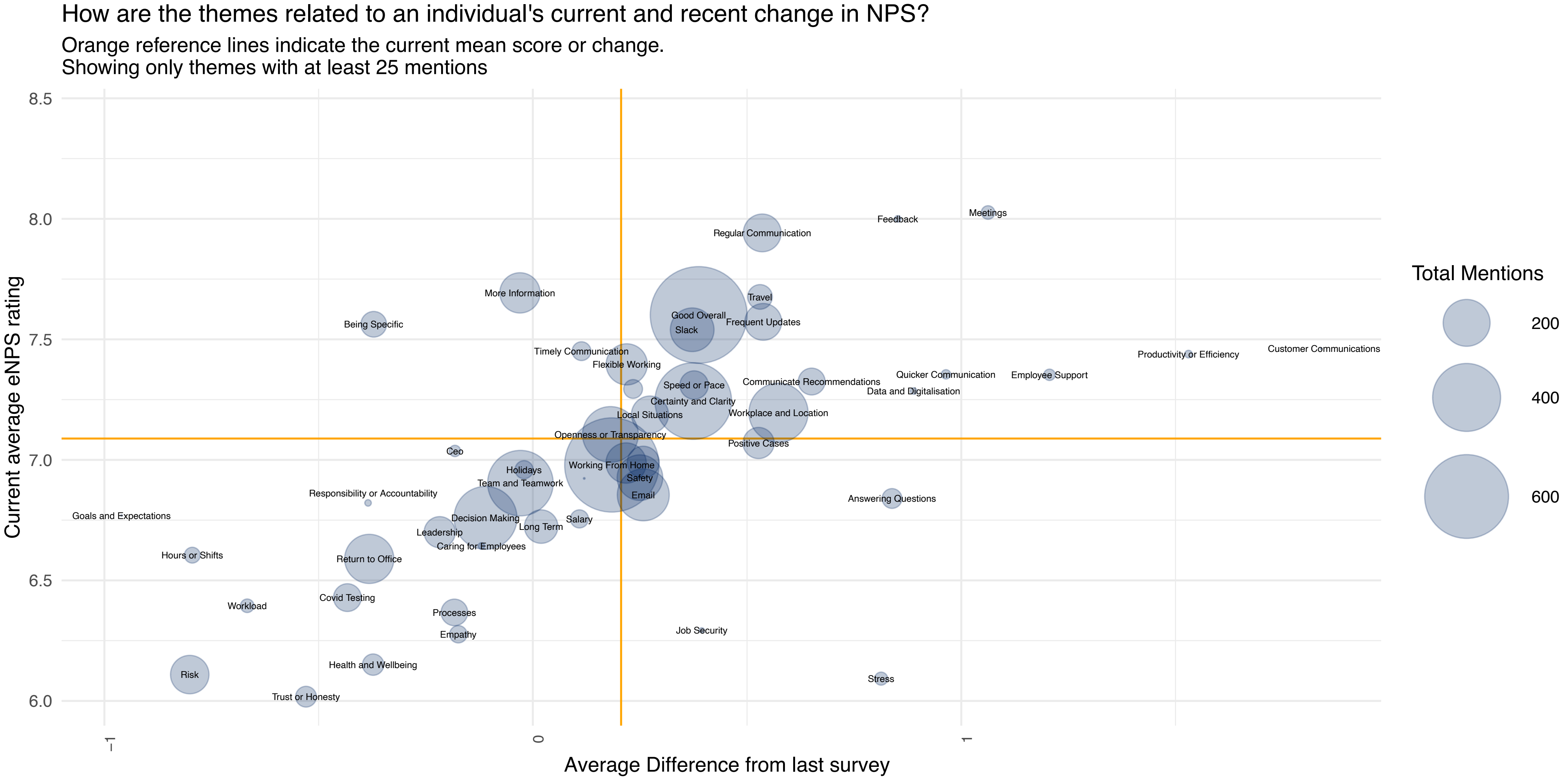
Text themes can be linked to scale questions

Using a key variable as an alternative to sentiment



Understanding changes in perspectives after events

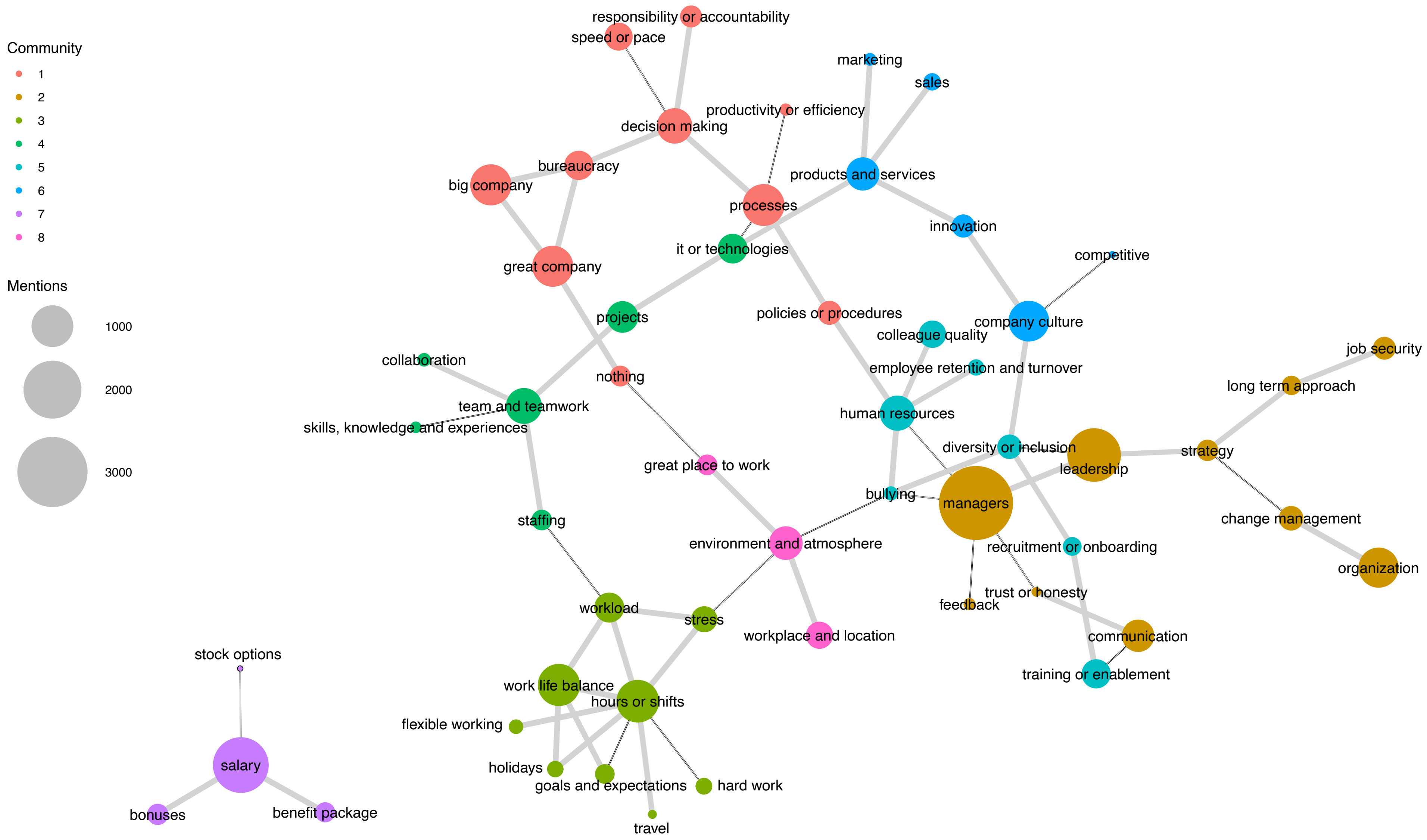
Longitudinal 'panel' data lets us explore topics associated with changes in a key metric.



Co-occurrence shows patterns in 'speech'

Network visualisation of topic cooccurence

Showing relationships where the probability of them being strongly related is at least 98%



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Co-occurrence shows patterns in ‘speech’

Statistical models identify probability themes are mentioned together. This question is about hybrid working.

Network visualisation of topic cooccurence

Showing relationships where the probability of them being strongly related is at least 85%